

20-S2-Q2

Solution (a) Heterogeneous Comprehensive-Learning PSO (HCLPSO)

Feature	Sub-population A – “global explorers”	Sub-population B – “local exploiters”
Size	$\approx 30 - 60$ % of swarm	Remaining particles
Learning rule	Comprehensive Learning (CLPSO)– each dimension d of particle i copies the d -th component of the personal-best of <i>some</i> neighbour chosen from the whole swarm (dimension-wise exemplar learning).	Canonical gBest PSO (or inertia-weight PSO)– velocity updated toward the particle’s own pBest and the global best.
Inertia weight w	Relatively large / time-varying (e.g. $w \approx 0.9 \rightarrow 0.4$) to broaden search.	Small / damped (e.g. $w \approx 0.6 \rightarrow 0.2$) to tighten convergence.
Acceleration coefficients	c_1 (high, 2.0–2.5) to stress self-exploration, c_2 (low, 0.5–1.0).	$c_1 \approx 1.0$, c_2 high (2.0–2.5) to follow elite.
Velocity clamp / constriction	Loose (large $V_{\max}$ or constriction $\chi \approx 1$)	Tight (smaller $V_{\max}$ or $\chi < 1$).
Adaptive mechanisms	• Dimension-level ‘refresh’ probability P_r— if a dimension has stagnated for T iterations, choose a new exemplar. • Periodic information migration: best p % of sub-pop A replace worst p % of sub-pop B to spread good directions.	• Periodic re-initialisation of worst particles to escape local minima (uses information coming from sub-pop A).

Workflow

1. Randomly split the swarm into the two groups once at start (or resample every K iterations).
2. Update each group with its dedicated equations/parameters.
3. Synchronise: share the global best **gBest** and perform migration.
4. Stop when a termination rule is met (max iter, target error, etc.).

The heterogeneity lets one part of the swarm keep global coverage while the other part refines local basins, giving better robustness than a uniform CLPSO or canonical PSO alone.

2 (b) NSGA-II survivor selection for a 6-member population

Step 1 Non-dominated sorting of the 12 combined solutions

Front	Members	Reason
F₁	S1 (1, 4), S2 (4, 1), S4 (3, 2)	None of them is dominated by any other solution.
F₂	S10 (6.5, 6.5), S11 (5.5, 7.5), S8 (5, 8), S5 (7, 6), S3 (6, 7), S7 (8, 5)	Each is dominated only by members of F ₁ .
F₃	S12 (9, 9)	Dominated by some in F ₂ but by none in lower fronts.
F₄	S6 (9.5, 9.5)	Dominated by F ₁ –F ₃ , not by S9.
F₅	S9 (10, 10)	Worst in both objectives.

(Dominance test: a solution dominates another if it is **no worse in both** objectives and **better in at least one**.)

Step 2 Fill the new population until the size limit (6)

- **Add all of F₁** – population now holds {S1, S2, S4}; 3 slots left.
- **Need 3 more from F₂** – use crowding distance.

Step 3 Crowding-distance calculation inside F₂

Compute distance separately for each objective (after sorting ascending because we minimise) and sum them; boundary solutions get ∞ .

Solution	Obj 1 rank	Obj 2 rank	Crowding distance
S8 (5, 8)	min $\rightarrow \infty$	max $\rightarrow \infty$	∞
S7 (8, 5)	max $\rightarrow \infty$	min $\rightarrow \infty$	∞
S5 (7, 6)	5	2	1.0
S10 (6.5, 6.5)	4	3	0.667
S11 (5.5, 7.5)	2	5	0.667
S3 (6, 7)	3	4	0.667

(Normalised distances: $(\text{next}-\text{prev})/(\text{max}-\text{min})$; details omitted for brevity.)

Step 4 Select the most widely spread members from F₂

Rank by descending crowding distance $\rightarrow$ S8, S7 (∞ each), then S5 (1.0).
Add them to the population.

Next-generation population

$\{S1, S2, S4, S8, S7, S5\}$
 $\boxed{\{S1, S2, S4, S8, S7, S5\}}$

These six individuals are the entire first front plus the three most widely spaced members from the second front, preserving both elitism and diversity as required by NSGA-II.